

## MILESTONE 2

M2 Low-fidelity Prototypen

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Travel Planner App

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## 1. Collection of ideas

**Methodology: Rapid Digital Iteration using SCAMPER** Instead of traditional paper sketching, our team decided to conduct the idea collection phase directly in code using basic HTML/CSS (Coded Wireframes).

**Process & Results:** By instantly changing HTML elements and CSS grid structures, we could immediately see how different ideas impacted the cognitive load.

- **Combine:** We experimented with combining multiple transport modes (e.g., taxi, train, and subway) into a single visual block rather than separate tabs.
- **Eliminate:** We tested eliminating traditional heavy filtering menus, replacing them with simple toggle switches for "Fastest Route" and "Eco-friendly".
- **Modify:** We modified the visual hierarchy by making total travel time and CO2 emissions the largest text elements on the screen. This rapid code-sketching allowed us to evaluate multiple structural ideas in real-time before finalizing the core concepts for our individual low-fi prototypes. (Note: Screenshots of these early, experimental HTML layout variations are attached below).

Travel Planner

[+] Mode: Business/Freelance

From: Vienna, Austria

To: Munich, Germany

Date: Monday, April 20

Preferences:

Fastest Route [X]

Minimal Transfers [X]

Business Class Only [ ]

FIND ROUTES

Fastest Options

FASTEST & MOST COMFORTABLE

08:00 - 10:15 2h 15m

Direct High-Speed Train (RJX)

Price: €89.00

DOOR-TO-DOOR

07:30 - 10:45 3h 15m

Taxi + Train + Subway

Price: €115.00

BUDGET OPTION

06:00 - 11:30 5h 30m

Bus (2 Transfers)

Trip Details

Vienna -> Munich

Arrival guaranteed before 11:00 AM meeting.

07:30 - Pickup  
Taxi from home address

08:00 - Departure  
Train Station (Platform 4)

10:15 - Arrival  
Munich Central Station

Total: €115.00

SEAMLESS ONE-CLICK BOOKING

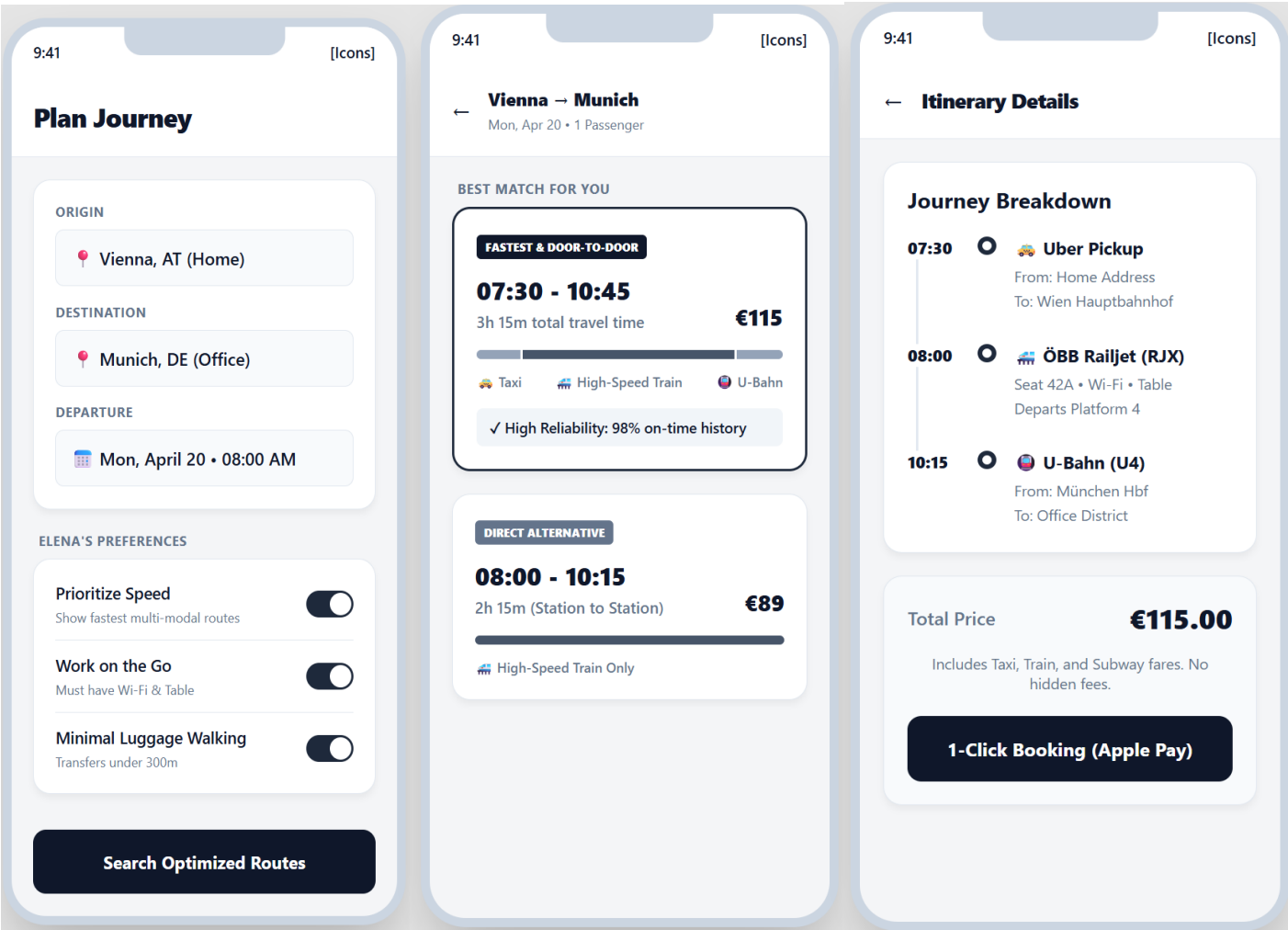
Includes Taxi, Train, and Local Transit tickets.

2. Low-Fidelity Prototype (Artur Ponomarenko)

For my specific prototype, I focused on **Primary Persona 2: Elena Rossi** (28, Freelance UX/UI Designer). Her main priorities are optimizing travel time, maximizing comfort, and minimizing transfers while working on the go.

**Prototyping Method: HTML Prototyping (Coded Wireframes)** - This method allowed me to design a clean, structured, and scrollable interface that accurately simulates a real mobile app's architecture. To ensure it remained strictly "low-fidelity" and focused purely on UX rather than UI, I used a black-and-white color palette, monospace typography, and basic geometric borders.

- **Screen 1 (Search & Preferences):** Elena values time over money. I added quick-toggle preferences directly on the search screen, such as "Fastest Route" and "Minimal Transfers," allowing her to instantly bypass budget options that take too long.
- **Screen 2 (Route Results):** To reduce cognitive load (a key issue identified in our literature research), the results are organized using distinct badges like "Fastest & Most Comfortable" The layout clearly emphasizes the total travel time (e.g., 2h 15m) rather than focusing heavily on price.
- **Screen 3 (Trip Details & Booking):** Elena’s major pain point is buying tickets across multiple platforms for a multimodal trip. I designed a highly structured timeline view of her transfers and added a prominent **"Seamless One-Click Booking"** button. This conceptually allows her to pay for the taxi, train, and local transit in a single action, eliminating booking friction.



### 3. Low-Fidelity Prototype (Anastasiia Trofymenko)

For my specific prototype, I focused on **Primary Persona 1: Lukas Weber (22, University Student)**. His main priorities are finding the most cost-effective routes that don't compromise his environmental values, while also managing the stress of complex intermodal transfers during his frequent trips home.

**Prototyping Method: HTML Prototyping (Coded Wireframes)** Following our team's digital-first approach, I developed an interactive HTML wireframe. This allowed me to test the "Priority Hub" logic—a core feature designed to reduce cognitive load by allowing the user to weigh their values before seeing the results. To maintain a low-fidelity aesthetic, I used a "hand-drawn" font style, dashed borders for input fields, and a grayscale palette to focus purely on the functional hierarchy.

- **Screen 1 (The Priority Hub):** Lukas often feels overwhelmed by long lists of results. I designed a "Pre-Filter Hub" featuring three primary sliders: **Budget, Speed, and Eco-Friendly**. Instead of digging through complex nested menus, Lukas can adjust these sliders to instantly re-rank journeys. This addresses the "Time Pressure" and "Choice Overload" identified in our literature research.
- **Screen 1&2 (Modular Journey Results):** To simplify intermodal travel, I replaced the traditional list view with a **Horizontal Modular Timeline**. Each leg of the trip (Bus, Train, Walk) is represented as a proportional block. This gives Lukas an immediate visual grasp of the journey's structure. I added a high-visibility **"ECO CHOICE"** badge to journeys with the lowest CO<sub>2</sub> impact, directly appealing to Lukas's environmental motivations.
- **Screen 2 (Stress-Aware Segment Details):** A major pain point for Lukas is "Transfer Anxiety." By clicking on a journey card, a detailed breakdown expands. I implemented **Stress-Level Labels** (e.g., "[STRESS-FREE]") that appear automatically if a transfer buffer exceeds 15 minutes. This provides the psychological reassurance Lukas needs to trust an intermodal route over a direct, more expensive car trip.

The image displays two wireframe screens for a travel planner application, labeled 'TravelPlanner v0.1'.

**Screen 1 (Left):** This screen features a search interface with 'From: Munich Central' and 'To: Prague Station' fields. Below these are three sliders for 'Adjust your priorities': BUDGET, SPEED, and ECO-FRIENDLY, each with a 'ANY' label. The 'Suggested Journeys' section shows two cards. The first card, labeled 'ECO CHOICE', shows a journey with a total time of 3h 20m and 1 transfer, priced at € 24.50. The second card, labeled 'FASTEST', shows a journey with a total time of 1h 15m and 0 transfers, priced at € 89.00. The bottom navigation bar includes 'SEARCH', 'SAVED', 'HISTORY', and 'SETTINGS'.

**Screen 2 (Right):** This screen shows a detailed breakdown of a selected journey. It includes a section 'Adjust your priorities' with sliders for BUDGET, SPEED, and ECO-FRIENDLY. Below this is the 'Suggested Journeys' section, which shows the 'ECO CHOICE' journey selected. The 'Journey Breakdown' section details the route: 08:00 - Bus Line 102 (45m), 15 min Transfer ([STRESS-FREE]), 09:00 - Railjet 543 (2h 10m), and 10 min Walk to destination. The total time is 3h 20m and the price is € 24.50. The bottom navigation bar includes 'SEARCH', 'SAVED', 'HISTORY', and 'SETTINGS'.

#### 4. Low-Fidelity Prototype (Oleh Shynkaryk)

For my prototype project I chose to create a product for Camille Dubois (age 23, fashion marketing intern) who has indicated that her three main goals when it comes to trip planning are simplicity, speed and minimal effort; she prefers to rely on pre-existing options (not do it yourself) and often feels overwhelmed by the number of possibilities that exist.

##### Prototype creation:

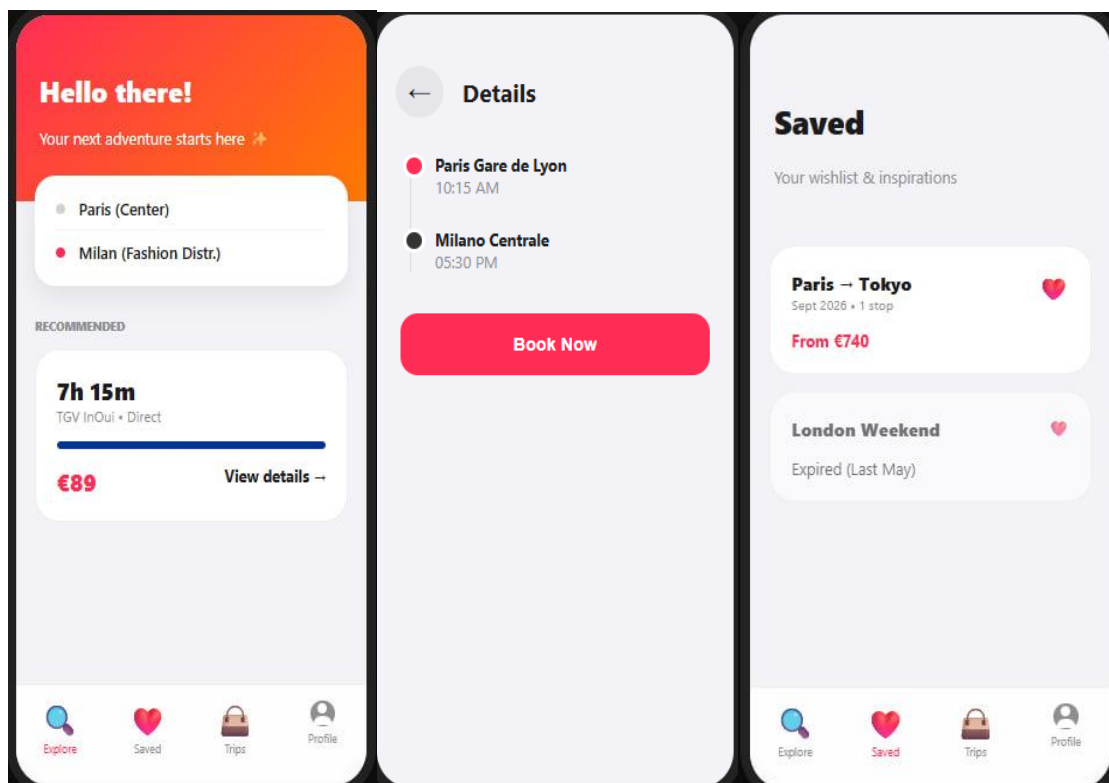
I created a prototype using a low fidelity mobile-first approach with a simple layout and easy-to-use navigation. My focus was on developing a user-friendly and rapid interaction experience rather than providing a high level of detail in the visuals.

**Screen 1: Explore (Main menu):** Camille will see a limited set of recommended routes that she can choose from (as opposed to an overwhelming number). Each route will utilize a card format that allows for easy identification of critical information (time and cost) thus reducing cognitive load and enabling Camille to quickly select from options that are appropriate for her.

**Screen 2: Route Information/Details:** The only information presented to Camille for booking a trip will be essential information for booking (departure time and arrival time). A clearly labeled "Book now" button will provide an easy way for Camille to reserve a trip quickly.

**Screen 3: Saved trips:** Camille will be able to collect and save ideas for future travels. The screen will have a simple layout with easy-to-scan information to help Camille with her habit of collecting travel ideas.

**Overall prototype design:** Camille's needs were addressed by reducing complexity of the trip planning process by limiting the number of trip options available, avoiding complex filtering systems and providing a quick and easy means of interaction.



## 5. Low-Fidelity Prototype (Danylo Prokhorenko)

For my prototype, I focused on our secondary persona: **Klaus Ziegler (68, retired teacher)**. His main priorities are the most stressfree and simple travel planning possible, without information overload. Although the app combines different modes of transport and routes (a core feature of our Travel Planner), this complexity must remain invisible in the background for Klaus.

### **Prototyping Method:**

HTML Prototyping (Coded Wireframes) – In line with our team's digital approach, I developed an interactive wireframe. To minimize the cognitive load for Klaus, the design was kept extremely simple (high-contrast, black and white, large typography). Complex filters, like those found with Persona 1 or 2, were completely omitted. I also used German as an interface language since there is a high chance, that Klaus doesn't understand English

### **Screen 1 (Focus Search):**

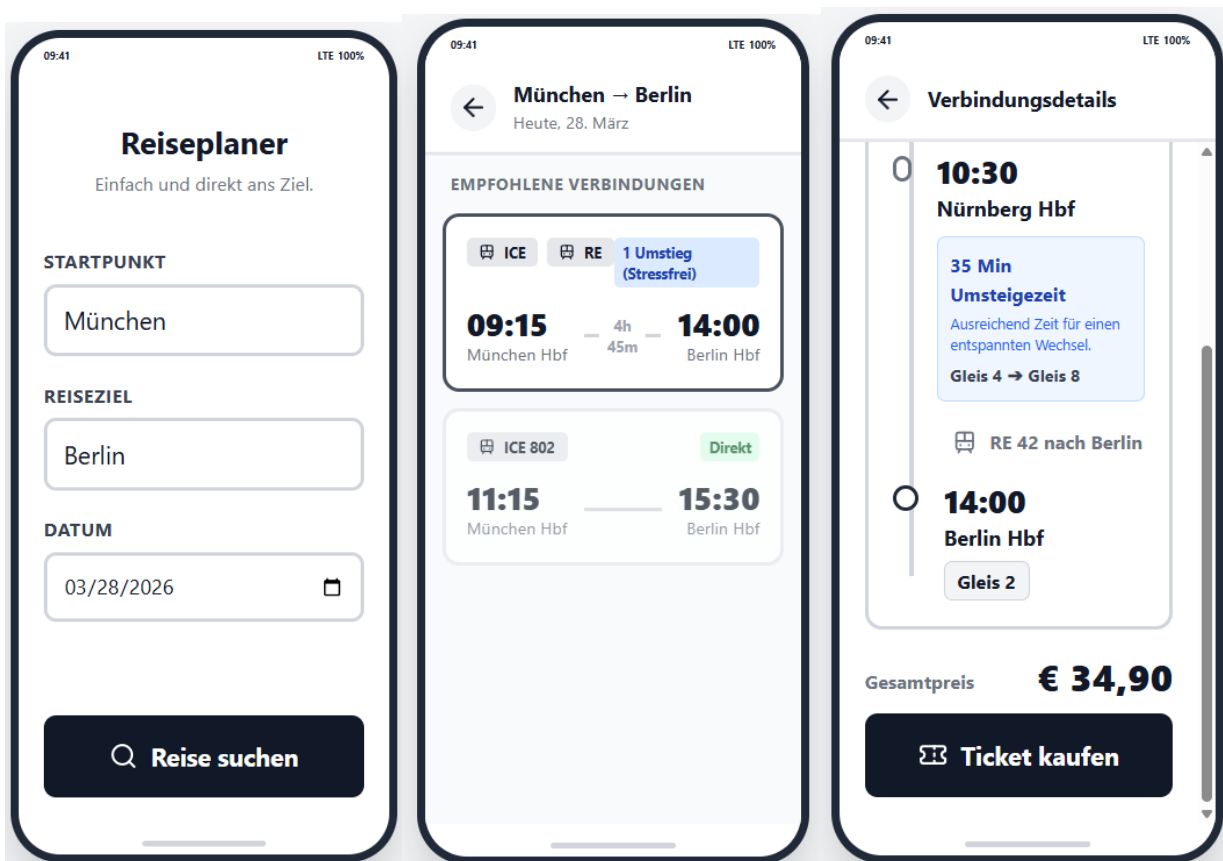
Klaus is often overwhelmed by apps with too many buttons. Therefore, the home screen consists of only the absolute minimum: huge, easy-to-read input fields for start and destination, as well as an unmissable "Search Journey" button. Confusing filters for CO2 emissions or multimodality are hidden by default.

### **Screen 2 (Curated Results):**

Instead of a confusing list of fragmented modes of transport, the app shows pre-combined, logical routes (e.g., ICE + regional train). To alleviate Klaus's fear of transfers, the information is greatly reduced and psychologically framed positively (clear badge "1 Transfer - Stress-free"). The app automatically links the means of transport for him.

### **Screen 3 (Simple Details & Guided Transfer):**

Klaus plans his day in his head and does not need exact pedestrian navigation, but rather reliability. The detail screen visualizes the transfer very transparently: It highlights the generous transfer time (e.g., 35 minutes) as a separate block and shows only the essential data such as platform changes. This simulates a guided, smooth process and solves his core problem of transfer anxiety.



## 6. Evaluation of Prototype 1 (Anastasiia Trofymenko)

**Test User:** Yasya (20 years old). My friend, a Statistics student. She travels occasionally, mostly for visiting friends or short trips, and prefers simple and clear apps without too much complexity.

### Methodology:

We had an in-person conversation where I showed her screenshots and explained the logic of all four prototypes. I asked her to go through each one and describe her first impressions, usability, and what she liked or found confusing. After that, she rated each prototype on a 10-point scale.

### Results & Feedback:

#### Prototype 1:

Yasya found the idea of sliders for Budget, Speed, and Eco very intuitive and engaging. She said it feels “interactive and easy to understand” because she can immediately express her preferences without going through complex filters. However, she mentioned that she was not completely sure how strongly each slider affects the final result.

**Rating: 8.5/10**

#### Prototype 2:

She liked the clear structure and especially the “1-Click Booking” idea, saying it makes the app feel practical and time-saving. However, she felt that some of the filters and

labels might be confusing for casual users who are not familiar with travel planning.

**Rating: 8/10**

### **Prototype 3:**

This was her favorite prototype. She appreciated the clean and minimal design, saying that “it doesn’t overload me with information.” She liked that only a few route options are shown and that everything is easy to scan quickly. However, she mentioned that she might sometimes want more control or additional details.

**Rating: 9/10**

### **Prototype 4**

She understood that this prototype is designed for older users and appreciated its clarity. However, for her personally, it felt too basic and limited. She said she would prefer slightly more flexibility and options.

**Rating: 7/10**

Conclusion: Yasya preferred prototypes that balance **simplicity and clarity** with a small degree of control. She strongly disliked interfaces that require too much thinking or too many steps. The feedback confirms that reducing cognitive load is essential, but the app should still allow users to understand how their choices influence the results.

## 7. Evaluation of Prototype 2 (Artur Ponomarenko)

**Test User:** Danya (21 years old). My friend from Ukraine, studies Computer Science at KNU. He doesn't travel a lot, so he likes when apps are just simple and fast.

**Methodology:** We had a video call on Discord. I sent him the links and pictures of all 4 prototypes our team made. I asked him to click through them and tell me his thoughts out. After that, I asked him to rate every prototype on a 10-point scale.

### Results & Feedback:

Prototype 1 (Anastasiia - Sliders for students): Danya liked the idea with sliders for Budget and Eco-friendly, because it looks like a game. But he said that the result screen has a bit too much text and information blocks. Rating: 8/10

Prototype 2 (Artur - Freelancer focus): He liked this one Aswell. He said the "1-Click Booking" button is the best thing ever, because "buying tickets on three different websites is the main reason I hate traveling". But he didnt understood the "Transfers under 300m" filter. He asked: "Is 300 meters a long walk in a station or not?". Rating: 9/10

Prototype 3 (Oleh - Visual / Simple): Danya said this app looks very modern, like Instagram. It was very easy for him to understand because there are almost no filters, just "Book Now". But as a tech guy, he said he wants to see more options sometimes. Rating: 7/10

Prototype 4 (Danylo - Big text / Pensioner): Because the text was in German, he didn't understand the words, but he liked the huge input fields. He agreed that if he showed



this to his grandfather, it would be the only app he could actually use without asking for help. Rating: 8/10

Conclusion: Danya preferred the prototypes that save time. For normal people who don't travel every week, too many filters make the process scary. He said "1-Click Booking" is a must-have for our final app.

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## 8. Evaluation of Prototype 3 (Oleh Shynkaryk)

**Test User: Diana (19 years old).** My girlfriend, studies Computer Science at the University of Vienna. Travels occasionally for fun. Prefers apps that are visually clear and straightforward.

**Methodology:** Had an in-person interview. I presented to her screenshots of all 4 prototypes and asked her to comment on each of them. She should tell me her first impression and highlight what she likes and dislikes about each prototype. Then she evaluates all prototypes from 1 to 10.

### Results & Feedback:

#### **Prototype 1 (Anastasiia - Sliders):**

Diana said that the idea with sliders sounds interesting and interactive. The main disadvantage is the presence of a lot of text. It is difficult for the user to quickly understand what is happening here. Also, she thinks that the color palette used is not quite suitable. There are too many elements on one screen. **Rating: 7.5/10**

#### **Prototype 2 (Artur - Freelancer focus):**

Overall structure is good; the application seems functional and useful. Some of the filters are confusing. They have no effect on the result or their meaning is ambiguous. This requires more thought from the user. **Rating: 8/10**

#### **Prototype 3 (Oleh - Visual / Simple):**

It was the best among prototypes that Diana liked. She noted that the design is modern and elegant. Everything is simple and understandable for the user without any additional explanations. The interface does not require much effort from the user to get the desired results. However, she may occasionally need to view additional information. **Rating: 9/10**

#### **Prototype 4 (Danylo - Elderly users):**

Font size in this application is very large and readable. Overall, the app has a very clear interface. At the same time, she found it too simplified and somewhat limited in functionality. However, considering its target audience, this application works effectively. **Rating: 7.5/10**

**Conclusion:** Diana prefers applications with clear and simple interfaces, visually beautiful design, and few unnecessary options. She does not like applications where there is a lot of text or filters whose function is unclear.

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## 9. Evaluation of Prototypes 4 (Danylo Prokhorenko)

**Test User:** Kyrylo (22 years old). My friend from Ukraine, studies Computer Science at Kharkiv National University. He doesn't travel a lot, but sometimes he takes long trips.

**Methodology:** We had a conversation on Telegram. I sent him pictures of all 4 prototypes our team made. He recorded voice messages with his reactions to them, and then rated each prototype on a 10-point scale.

### Results & Feedback:

**Prototype 1 (Artur - Freelancer focus):** Kyrylo liked the modern design and the mix of white, gray, and black text with colorful emojis. However, he felt it was a bit overwhelming for what should be a simple app. He noted that filters aren't hidden behind a button and there is a lot of text. He also mentioned that showing details like a chosen seat, Wi-Fi, and platform before a person has actually booked doesn't make much sense. Rating: **7.5/10**

**Prototype 2 (Anastasiia - Sliders for students):** He liked the fresh idea of using sliders to pick priorities for budget, speed, and eco-friendliness. Since he currently lives in Germany, he compared it to the Deutsche Bahn interface, noting that this app handles more complex, worldwide travel (like planes) without making the interface much harder to understand. He did point out a lack of colors, finding the view a bit too simple and gray. Rating: **7/10**

**Prototype 3 (Oleh - Visual / Simple):** This was his favorite. He loved how colorful and yellow-looking the app is, praising its simplicity, readability, and youthful feel. He pointed out that there is almost no demonstration of how different transport modes (airplanes, trains, buses) are mixed. He also felt the bottom icons were a bit too childish, but overall he really liked it. Rating: **8.5/10**

**Prototype 4 (Danylo - Big text / Pensioner):** He liked this one the least because it felt like it was made for older people, and he doesn't feel old. He found the text too big and the interface too simplistic since it doesn't allow for custom setup. He also disliked the lack of colors and the absence of a menu. Rating: **6/10**

**Conclusion:** Kyrylo preferred designs that are easy on the eyes and youthful, striking a balance between simplicity and visual appeal. He disliked interfaces that are either too cluttered with early information or too restrictive and simplistic.

## 10. Summary of the Interviews with the Users final insight

As it can be seen from the results obtained during all the interviews, there is a distinct trend – most of the users liked those prototypes that had clear structure, quick action, and minimalism but lacked excessive texts, numerous options, or confusing filters. They desired to take decisions fast and without spending unnecessary time analyzing the interface of the system.

In order to make the comparison easier, let us determine the average rating for each prototype:

**Prototype 1** (Anastasiia) – 7.9/10 points

**Prototype 2** (Artur) – 8.0/10 points

**Prototype 3** (Oleh) – 8.4/10 points

**Prototype 4** (Danylo) – 7.1/10 points

**Prototype 3** by Oleh proved to be the winner since all users were positive about the modern look, neat layout, and clearness of this prototype. All these factors helped the users to analyze the interface easily and take appropriate decisions in a few seconds. However, some of the users noted that more information would not be superfluous, and one more option could be added without making the prototype complicated.

Another interesting prototype was **Prototype 2** by Artur since it had great functionality and such useful tools as "1-Click Booking" mentioned several times by the participants of the interview. On the other hand, some filters confused the users, and they did not like their complexity

**Prototype 1** created by Anastasiia was quite innovative because it provided interactivity through the use of sliders, etc. However, many of the users stated that this prototype contained too much information and too many texts, which complicated the analysis.

The worst prototype was **Prototype 4** by Danylo who managed to create something simple and clear with large text and easy to comprehend elements. However, most participants felt that this prototype is extremely limited and cannot be used as the only one since it lacks flexibility for the needs of various users.

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### Final insight:

The findings of our research have shown that the best solution will include:

- Simplicity and clearness of **Prototype 3**;
- Functionality and quickness of actions of **Prototype 2**;
- Limited but understandable customization options of **Prototype 1**.

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## 11. Work Distribution

**Anastasiia Trofymenko:** Project management aspects, creation of the Low-Fi prototype for Persona 1, and final document integration.

**Artur Ponomarenko:** Creation of the Low-Fi prototype for Persona 2, and conducted the evaluation interview with Test User 2.

**Oleh Shynkaryk:** Creation of the Low-Fi prototype for Persona 3, and conducted the evaluation interview with Test User 3.

**Danylo Prokhorenko:** Creation of the Low-Fi prototype for the Secondary/Negative Persona, and conducted the evaluation interview with Test User 4.

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## 12. Transparency on AI Usage

During the preparation of this milestone, we used AI (Gemini) as a supportive tool to optimize our workflow in the following ways:

1. **Assistance in Prototyping:** Fixing errors in HTML and CSS to facilitate rapid "coded wireframing" and digital iteration during the ideation phase.
  2. **Copywriting & Formatting:** Translating and structuring the final report text to ensure professional and grammatically correct academic English.
  3. **Ideation Support:** Assisting in brainstorming specific UI feature adaptations based on the persona descriptions defined in Milestone 1.
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